Assignment - AI

Course: 696a81e602c34352d7e9f7d9 | Due Date: 1/23/2026
Questions: 5 | Total Marks: 50
Type: Descriptive Questions

QUESTIONS:

Q1. Explain the concept of machine learning in the context of AI. Provide examples of how machine learning is used in real-world applications.

[10 Marks | Difficulty: medium]

Expected Points:
- Definition of machine learning in AI
- Examples of real-world applications (e.g., recommendation systems, image recognition)

Q2. Discuss the difference between narrow AI and general AI. Provide examples to illustrate each type of AI.

[10 Marks | Difficulty: medium]

Expected Points:
- Definition and characteristics of narrow AI
- Definition and characteristics of general AI
- Examples of narrow AI (e.g., virtual assistants)
- Examples of general AI (e.g., human-level intelligence)

Q3. Explain the importance of data preprocessing in AI. Describe at least three common techniques used in data preprocessing.

[10 Marks | Difficulty: medium]

Expected Points:
- Significance of data preprocessing in AI
- Explanation of techniques (e.g., normalization, feature scaling, outlier detection)

Q4. Describe the concept of neural networks in AI. Explain the role of activation functions in neural networks and provide examples of popular activation functions.

[10 Marks | Difficulty: medium]

Expected Points:
- Definition of neural networks in AI
- Explanation of activation functions and their role
- Examples of popular activation functions (e.g., ReLU, Sigmoid)

Q5. Discuss the ethical considerations in AI development and deployment. Provide examples of potential ethical issues that may arise in AI applications.

[10 Marks | Difficulty: medium]

Expected Points:
- Explanation of ethical considerations in AI
- Examples of ethical issues (e.g., bias in algorithms, privacy concerns)